QUANTUM MEMORY

Room Temperature Rb-based Quantum Memory Experimentation Kit for Academia

This kit is a true Quantum Optics experimental platform designed for Generation, Storage, and Retrieval of Photonic pulses using Rubidium Vapor-based Quantum Memory, functional at room temperature. Equipped with Quantum Storage, Detection and a Laser Input Control System, it operates on Electromagnetically Induced Transparency (EIT) principles. This product consists of two modules :

I. Quantum Memory Experimentation
II. Input Laser Control System

I. Product Description of Quantum Memory Experimentation Kit

The first module of the Quantum Memory Experimentation Kit showcases the storage and retrieval processes of Photonic Polarization Pulses in a Rubidium (Rb)-based hot vapor ensemble. It facilitates a range of experiments related to quantum storage and characterization. The design of the quantum storage unit includes shielding to protect it from external magnetic fields.

Key Features

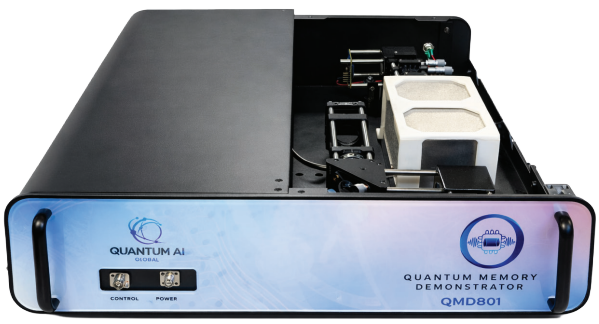
- Photonic Polarisation Pulse Storage
- Room Temperature Operation
- Long Storage Time
- Long-lasting Alignment Free Operation
- Resistant to Electromagnetic Noise
- Upgradable to Single Photon Level Quantum Memory

Product Specifications

Parameter	Unit	Min.	Max.	Typical
Operating Wavelength	nm	-	-	794.8 +- 0.1
Pulse Duration	µs	0.2	10	4
Storage Time	µs	0.2	20	8
Efficiency	%	-	-	10
Probe Input Power	mW	-	-	2
Control Input Power	mW	-	-	5
Operating Temperature of Vapour Cell	°C	30	110	55
Ambient Temperature	°C	10	45	20
Warm-up Time	min	-	-	15
Dimensions	mm^3	-	-	432*482*105
Weight	kg	-	-	15

Inputs/Outputs

Optical Inputs	2 x (FC/APC, PMF)
Analog Outputs	2 x BNC
Interface	USB



Quantum Memory Experimentation Kit

II. Product Description of Input Laser Control System (Pulse Module of Quantum Memory)

The second module of the Quantum Memory Kit, the Input Laser Control System, is a versatile solution for precise Signal Modulation and Control. This system is designed to generate two laser output signals separated by a 6.8 GHz frequency difference for Rubidium-based Quantum Memory operations. It is capable of generating laser pulse sequences with flexible parameter programming through an intuitive GUI. However, the techical parameters can be adjusted to other frequency differences for a variety of atomic physics experiments, including those involving three-level systems such as EIT, CPT, and cold atom-based applications.



Input Laser Control System

Product Specifications

Parameter	Unit	Min.	Max.	Typical
Operating Wavelength	nm	-	-	794.8 +- 0.1
Linewidth	MHz	-	-	<6
Continuous Adjustment Free Duration	Hrs	-	-	10
Control Laser Output Power	mW	-	-	10
Probe Laser Output Power	mW	-	-	10
Warm-up Time	mins	-	-	30
Ambient Temperature	°C	10	45	20
Dimensions	mm^3	-	-	432*482*105
Weight	kg	-	-	12
Power Consumption	W	180	-	180

Inputs/Outputs

Optical Outputs	2 × (FC-APC,PMF)
Analog Inputs	1 × SMA
Analog Outputs	1 × SMA
Interface	Ethernet and USB

Key Features

- Room Temperature Operation
- Intuitive Software Interface
- Pulse Shaping Control
- Long-lasting Alignment Free Operation

Experimentation Outcomes

The Quantum Memory Demonstrator provides students, researchers, and professionals with a true and robust Quantum platform for experimentation. The aforementioned setup can be used for the following applications:

- Electromagnetically Induced Transparency (EIT) Demonstration
- EIT Linewidth Measurement
- Storage and Retrieval of TTL pulses
- Retrieval Efficiency Calculation
- Storage time optimization
- Storage time vs Efficiency measurement